MYOUNGCHUL KIM

Data Scientist

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ABOUT ME

Data Scientist with 10+ years of **statistical modeling** *and* **large-scale data analysis** *(∼200 TB) across* **AstroParticle Physics** *research and industry. Experienced in* **LLM multi-agent systems** (**LangGraph**, **RAG**), **prompt engineering**, **Bayesian** *and* **GNN** *modeling*, **SFT**/**RLHF**, *and* **machine learning** *risk assessments on Dagster* **MLOps** *pipelines (Python, AWS).*

TECHNICAL SKILLS

Coding Tools: Python | C++ | Git / GitHub | Vim | Jupyter | Latex | ROOT | HTCondor | CVMFS

Data Analytics: NumPy | SciPy | Scikit-learn | Pandas | GeoPandas | SQL | Matplotlib | Seaborn

Modelling: TensorFlow | PyTorch | PyTorch Geometric | XGBoost | Unsupervised Learning | CNN | GNN | Time series | Ensemble Methods | Statsmodels | Stan | Bayesian Inference | LLM Agent | RAG | Prompt Engineering | RLHF | SFT | Hugging Face

Deployment: AWS | GCP | Docker | Kubernetes | Databricks | Dagster | LangChain | LangGraph | FastAPI | Streamlit | Raspberry Pi

EXPERIENCE

Research Engineer

DataLabs

August 2026 – Ongoing Tokyo, Japan

LLM Agents and 3D Point-Cloud R&D for Construction Products

August 2026 – Ongoing Role: Research Engineer Team size: 4

- Build an **LLM agent** that incorporates complex rule-based math, bringing **AI** capabilities into **BIM/CIM** construction products.
- Research core **3D data processing** algorithms for **point clouds** (LiDAR) and BIM/CIM models to improve existing products and create new ones.
- Run client **PoCs** with construction and infrastructure partners, from requirements to proposals and reports written in **Japanese**.

Data Scientist

Fracta

November 2024 – July 2026 Tokyo, Japan

LLM-Powered Multi-Agent System for Raw Data Preprocessing

March 2026 – July 2026 Role: PoC Lead Team size: 2

- Built a **LangGraph multi-agent** PoC (**Claude**, **GPT-4o**, **Gemini**) that turns messy client GIS and tabular data into schema-validated **ML** inputs.
- Cut **token usage** 50–60% by splitting column and value mapping into two **LLM** agents; added **prompt caching** for address and date cleaners.
- Ensured reliability via in-prompt docs (chosen over **RAG**), **Pydantic**/rule-based validation with LLM retries, and 14 **human-in-the-loop** checkpoints.

- Contributed to rebuilding the PoC for production as a **Claude Code** orchestrator with 4 **subagents**, validated by the **Dagster** ETL's **Pandera** schemas.

AI-Driven Likelihood-of-Failure (LOF) Risk Assessments for Water Utilities

November 2024 – July 2026 | Role: Data Scientist | Team size: 4

- Generated LOF predictions with **XGBoost** models (global, local, hybrid) trained on $\sim 10^5$ km of pipes from $\sim 10^2$ utilities, with high held-out **AUC**.
- Worked with the multi-utility model, retrained via distributed **XGBoost** on **Databricks** with models logged to **MLflow**.
- Selected the best model per client using **cumulative gain** (break capture among the highest-risk pipes) on **time-based validation**.
- Delivered **risk assessments** to government water utilities, from raw-data cleaning to LOF rankings, **SHAP** risk drivers, and long-term **survival** forecasts.

End-to-End MLOps Pipeline (Dagster on AWS EKS)

November 2024 – July 2026 | Role: Data Scientist | Team size: 4

- Engineered an 8-stage **Dagster** pipeline automating ingestion, cleaning, imputation, modeling, survival analysis, and client report generation.
- Ran the pipeline on **AWS EKS** (**Docker**, **Kubernetes**), using a versioned run launcher to rerun any released pipeline version.
- Automated **CI/CD** with **GitHub Actions**: Ruff and pytest on pull requests, **ECR** image builds, and rollouts to EKS on merge.

Bayesian & GNN Missing-Data Imputation for Pipe Networks

November 2024 – May 2026 | Role: PoC Lead | Team size: 2

- Integrated a **Bayesian** imputer (**Stan**; variational inference + **MCMC**) into the **Dagster** pipeline to fill missing install year, diameter, and material.
- Audited 8 **Stan** models and fixed 17 issues, including silent data corruption, numerical crashes, non-reproducible runs, and cloned **MCMC** chains.
- Rewrote the neighbor likelihood as a **Bayesian model averaging** mixture, enabled an **empirical-Bayes** material prior, and removed a biased fallback.
- Built a **GNN** imputer in **PyTorch Geometric** (adversarial GCN autoencoder, **WGAN-GP**) with **uncertainty quantification** via MC dropout and **VGAE**.

Data Science Instructor

Le Wagon

October 2024 – Ongoing | Tokyo, Japan

Data Science & AI Bootcamp Instruction

October 2024 – Ongoing | Role: Teaching Assistant | Team size: 13

- Instructed **Python** for Data Science, covering data extraction, manipulation, visualization, **and statistics**.
- Guided students through **ML/DL** workflows using **Scikit-Learn** and **neural network** architecture design.
- Delivered training in **ML Engineering**, including **Python** package deployment on **GCP** and ethical implications of **AI**.

Python Data Scientist/Analyst

Turing

July 2024 – January 2025 | Remote

Multilingual SFT and RLHF Implementation for Next-Generation AI

July 2024 – January 2025 | Role: Data Scientist | Team size: 6

- Performed **Supervised Fine-Tuning (SFT)** and **Reinforcement Learning with Human Feedback (RLHF)** across **English, Korean**, and **Japanese** datasets to improve **LLM** alignment.
- Analyzed model performance via **Loss Pattern** tracking and **Output-Based Evaluation** to enhance robustness and reliability.

- Curated model responses through **Comparative Analysis** and **Preference Pair** evaluation, ensuring contextually accurate outputs.

AI Training for Japanese Writers

Outlier

July 2024 – October 2024 | Remote

Fine-Tuning and Validation of AI-Generated Japanese Voice

July 2024 – October 2024 | Role: LLM Evaluator | Team size: 12

- Validated **AI-generated voice** outputs, assessing quality via **data augmentation, pitch**, and **speech rate** for natural and contextually accurate results.
- Applied **linguistic expertise** in Japanese to evaluate **voice model** performance against clarity, consistency, and **cultural accuracy** benchmarks.

Graduate Research Assistant, Ph.D.

International Center for Hadron Astrophysics (ICEHAP)

April 2017 – December 2023 | Chiba University, Japan

Search for Ultra-high Energy Neutrinos from Askaryan Radio Array (ARA) by Template Method

April 2018 – December 2023 | Role: Project Lead | Team size: 6

- Classified astronomical signals using **PCA** on ∼200 TB of **radio-frequency** data, optimized via **Frequentist Statistics** and **Pseudo Experiments**.
- Implemented **FFT**, **Interferometry**, and **Matched Filter** for feature extraction on large-scale **CPU & GPU** clusters using scientific **Python** and **C++**.
- Evaluated results via **statistical significance**, **systematic uncertainty**, and **Monte Carlo simulation**; pipeline adopted by international collaborators.

Development of In Situ Antenna Model for Simulation

April 2017 – March 2019 | Role: Data Management | Team size: 4

- Performed high-precision calibration for **radio-frequency antennas** to measure and characterize physical properties.
- Extracted feature patterns from raw data through **noise & signal analysis**, contributing to simulation model development.

Graduate Research Assistant, MSc.

Neutrino AstroParticle Physics Lab (NAPPL)

March 2015 – February 2017 | SungKyunKwan University, South Korea

IceCube Camera System to Study Properties of the Antarctic Ice

March 2015 – February 2017 | Role: Project Lead | Team size: 5

- Classified intrinsic camera noise under extreme low-temperature conditions using **feature extraction** and **image analysis**.
- Developed **Python** package for camera control via **Raspberry Pi**, enabling automated data collection.
- Evaluated camera system performance by applying **statistical techniques** to large-scale **image datasets**.

PROJECT

AI Music Generation via RNN

Le Wagon Data Science & AI Bootcamp

January 2024 – March 2024 | Le Wagon, Tokyo

Role: Data Management | Team size: 4

- Architected **RNN** to learn musical patterns from large classical music datasets encoded in numerical form, enabling customizable music generation.
- Deployed end-to-end pipeline via **FastAPI** and **Streamlit**, integrating generated outputs with **Ableton**.

- Managed data preprocessing and model training workflows, ensuring scalable and reproducible **deep learning** pipelines.

EDUCATION

Data Science & AI Bootcamp
Le Wagon

January 2024 – March 2024 | Japan

- Studied **Python** for Data Science, covering **data extraction**, **manipulation**, **visualization**, **statistics**, and **linear algebra**.
- Built end-to-end **ML/DL** workflows using **Scikit-Learn** and designed **neural network** architectures for practical applications.
- Developed **Python** packages for large-scale data tasks on **GCP**, with exposure to **ML Engineering** and **AI ethics**.

Completion of Ph.D. program (ABD), AstroParticle Physics
Chiba University

April 2017 – December 2023 | Japan

- Research topic: Search for Ultra-high Energy Neutrinos Using Eight Years of Data from Two ARA Stations by the Neutrino Template Method

Master of Science, AstroParticle Physics
SungKyunKwan University

March 2015 – February 2017 | South Korea

- Thesis title: Performance study of camera system for the IceCube-Gen2 detector

Bachelor in Science, Physics
SungKyunKwan University

March 2011 – February 2015 | South Korea

SELECTED PUBLICATIONS

Journal Articles

- P. Dasgupta, M. S. Muzio, et al., "Progress Towards a Diffuse Neutrino Search in the Full Livetime of the Askaryan Radio Array," *PoS*, vol. ICRC, p. 1226, 2023.
- M. Kim et al., "Enhanced Ultra-High Energy Neutrino Search at the Askaryan Radio Array using Template-based Techniques," *PoS*, vol. ICRC, p. 1148, 2023.
- D. Bose, M. Jeong, K. Woosik, J. Kim, M. Kim, C. Rott, et al., "PINGU camera," *PoS*, vol. ICRC, p. 1145, 2015.

AWARDS

Japanese Government Monbukagakusho Scholarship (MEXT)
Graduate Research Assistant in Ph.D., 2017 – 2020

Teaching Assistant (T.A.) of Korea & Japan Joint Government Scholarship
Teaching Assistant for a freshman Korean students, 2017 – 2018

BK21+ Research student scholarship
Graduate Research Assistant in Msc., 2015 – 2017

Operating Assistant scholarship
Physics Experiment Assistant, 2016 – 2017

CK Research student scholarship
Research Assistant in Bsc., 2014

INTERESTS

Classical Music | Orchestra | Contrabass | Universe | Fourier Transform